E- COMMERCE CAPSTONE PROJECT

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import json
import csv

# ------------------------------
# FUNCTIONS
# ------------------------------

def load_visits(filename):
    with open(filename, "r") as f:
        return [line.strip() for line in f]


def load_products(filename):
    with open(filename, "r") as f:
        data = json.load(f)

    products = {}
    for p in data["products"]:
        products[p["product_id"]] = {
            "name": p["name"],
            "price": p["price"]
        }
    return products


def load_orders(filename):
    orders = []
    with open(filename, "r") as f:
        reader = csv.DictReader(f)
        for row in reader:
            orders.append({
                "order_id": int(row["order_id"]),
                "customer": row["customer"],
                "product_id": int(row["product_id"]),
                "quantity": int(row["quantity"])
            })
    return orders


# ------------------------------
# LOAD DATA
# ------------------------------

visits = load_visits("website_visits.txt")
products = load_products("products.json")
orders = load_orders("orders.csv")

# ------------------------------
# PART 1 - WEBSITE ANALYSIS
# ------------------------------

print("\n--- Website Analysis ---")


print("All Visitors:", visits)
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print("Total Visits:", len(visits))



unique_visitors = set(visits)
print("Unique Visitors:", unique_visitors)


visit_count = {}
for v in visits:
    visit_count[v] = visit_count.get(v, 0) + 1

print("Visit Count:", visit_count)


most_visitor = max(visit_count, key=visit_count.get)
print("Most Frequent Visitor:", most_visitor)


# -------------------------------
# PART 2 - PRODUCT ANALYSIS
# -------------------------------

print("\n--- Product Analysis ---")

for pid, info in products.items():
    print(f"{info['name']} → {info['price']}")

most_exp = max(products.values(), key=lambda x: x["price"])
least_exp = min(products.values(), key=lambda x: x["price"])

print("Most Expensive Product:", most_exp)
print("Least Expensive Product:", least_exp)


# -------------------------------
# PART 3 - ORDERS ANALYSIS
# -------------------------------

print("\n--- Orders Analysis ---")

for order in orders:
    print(order)

product_quantity = {}
customer_orders = {}

for order in orders:
    pid = order["product_id"]
    customer = order["customer"]

    product_quantity[pid] = product_quantity.get(pid, 0) + order["quantity"]
    customer_orders[customer] = customer_orders.get(customer, 0) + 1

print("Quantity per Product:", product_quantity)
print("Orders per Customer:", customer_orders)


# -------------------------------
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# PART 4 - SALES CALCULATION
# -------------------------------

print("\n--- Sales Calculation ---")

total_revenue = 0
product_revenue = {}

for order in orders:
    pid = order["product_id"]
    qty = order["quantity"]
    price = products[pid]["price"]

    revenue = price * qty
    total_revenue += revenue

    pname = products[pid]["name"]
    product_revenue[pname] = product_revenue.get(pname, 0) + revenue

print("Total Revenue:", total_revenue)
print("Revenue per Product:", product_revenue)

top_product = max(product_revenue, key=product_revenue.get)
print("Highest Selling Product:", top_product)


# -------------------------------
# PART 5 - CUSTOMER ANALYSIS
# -------------------------------

print("\n--- Customer Analysis ---")

customer_spending = {}

for order in orders:
    name = order["customer"]
    pid = order["product_id"]
    qty = order["quantity"]

    amount = products[pid]["price"] * qty
    customer_spending[name] = customer_spending.get(name, 0) + amount

print("Customer Spending:", customer_spending)

top_customer = max(customer_spending, key=customer_spending.get)
print("Top Customer:", top_customer)

high_spenders = [c for c in customer_spending if customer_spending[c] >
50000]
print("Customers spending > 50000:", high_spenders)


# -------------------------------
# PART 7 - TUPLE USAGE
# -------------------------------

product_tuples = [(p, product_revenue[p]) for p in product_revenue]
print("Product Revenue Tuples:", product_tuples)


# -------------------------------
```

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# PART 8 - REPORT GENERATION
# -------------------------------

with open("sales_report.txt", "w", encoding="utf-8") as f:
    f.write("E-Commerce Sales Report\n")
    f.write(f"Total Website Visits: {len(visits)}\n")
    f.write(f"Unique Visitors: {len(unique_visitors)}\n")
    f.write(f"Total Revenue: {total_revenue}\n\n")

    f.write(f"Top Customer: {top_customer}\n\n")

    f.write("Product Sales\n")
    for p in product_revenue:
        f.write(f"{p} -> {product_revenue[p]}\n")


# -------------------------------
# FINAL CHALLENGE
# -------------------------------

print("\n--- Final Challenge ---")

order_customers = set([o["customer"] for o in orders])

# Visitors who never ordered
never_ordered = unique_visitors - order_customers
print("Visited but never ordered:", never_ordered)

# Customers who visited only once but ordered
one_time_visitors = [v for v in visit_count if visit_count[v] == 1]
special_customers = [c for c in customer_orders if c in one_time_visitors]

print("Ordered but visited once:", special_customers)
```
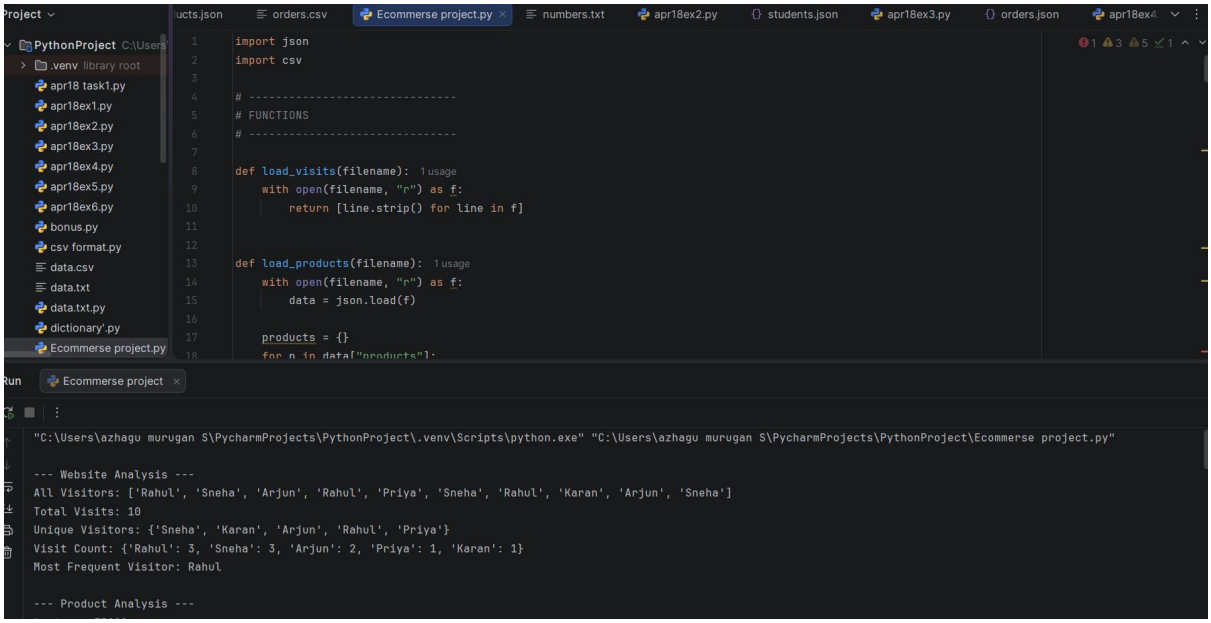

EXECUTION



```python

visits = load_visits("website_visits.txt")
products = load_products("products.json")
orders = load_orders("orders.csv")

# ------------------------------
# PART 1 - WEBSITE ANALYSIS
# ------------------------------

print("\n--- Website Analysis ---")

print("All Visitors:", visits)
print("Total Visits:", len(visits))

unique_visitors = set(visits)
print("Unique Visitors:", unique_visitors)

visit_count = {}
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\Ecommerse project.py"

--- Website Analysis ---
All Visitors: ['Rahul', 'Sneha', 'Arjun', 'Rahul', 'Priya', 'Sneha', 'Rahul', 'Karan', 'Arjun', 'Sneha']
Total Visits: 10
Unique Visitors: {'Sneha', 'Karan', 'Arjun', 'Rahul', 'Priya'}
Visit Count: {'Rahul': 3, 'Sneha': 3, 'Arjun': 2, 'Priya': 1, 'Karan': 1}
Most Frequent Visitor: Rahul
```

```python
# ------------------------------
# PART 2 - PRODUCT ANALYSIS
# ------------------------------

print("\n--- Product Analysis ---")

for pid, info in products.items():
    print(f"{info['name']} → {info['price']}")

most_exp = max(products.values(), key=lambda x: x["price"])
least_exp = min(products.values(), key=lambda x: x["price"])

print("Most Expensive Product:", most_exp)
print("Least Expensive Product:", least_exp)


# ------------------------------
# PART 3 - ORDERS ANALYSIS
```

```
Visit Count: {'Rahul': 3, 'Sneha': 3, 'Arjun': 2, 'Priya': 1, 'Karan': 1}
Most Frequent Visitor: Rahul

--- Product Analysis ---
Laptop → 75000
Mouse → 500
Keyboard → 1500
Monitor → 12000
Most Expensive Product: {'name': 'Laptop', 'price': 75000}
Least Expensive Product: {'name': 'Mouse', 'price': 500}
```

```python
# ------------------------------
# PART 3 - ORDERS ANALYSIS
# ------------------------------

print("\n--- Orders Analysis ---")

for order in orders:
    print(order)

product_quantity = {}
customer_orders = {}

for order in orders:
    pid = order["product_id"]
    customer = order["customer"]

    product_quantity[pid] = product_quantity.get(pid, default: 0) + order["quantity"]
```

```
Least Expensive Product: {'name': 'Mouse', 'price': 500}

--- Orders Analysis ---
{'order_id': 1, 'customer': 'Rahul', 'product_id': 101, 'quantity': 1}
{'order_id': 2, 'customer': 'Sneha', 'product_id': 102, 'quantity': 2}
{'order_id': 3, 'customer': 'Arjun', 'product_id': 103, 'quantity': 1}
{'order_id': 4, 'customer': 'Rahul', 'product_id': 102, 'quantity': 3}
{'order_id': 5, 'customer': 'Priya', 'product_id': 104, 'quantity': 1}
{'order_id': 6, 'customer': 'Sneha', 'product_id': 101, 'quantity': 1}
{'order_id': 7, 'customer': 'Karan', 'product_id': 103, 'quantity': 2}
```

```python
# ------------------------------
# PART 4 - SALES CALCULATION
# ------------------------------

print("\n--- Sales Calculation ---")

total_revenue = 0
product_revenue = {}

for order in orders:
    pid = order["product_id"]
    qty = order["quantity"]
    price = products[pid]["price"]

    revenue = price * qty
    total_revenue += revenue

    pname = products[pid]["name"]
```

```
Orders per Customer: {'Rahul': 3, 'Sneha': 2, 'Arjun': 1, 'Priya': 1, 'Karan': 1}

--- Sales Calculation ---
Total Revenue: 181000
Revenue per Product: {'Laptop': 150000, 'Mouse': 2500, 'Keyboard': 4500, 'Monitor': 24000}
Highest Selling Product: Laptop

--- Customer Analysis ---
Customer Spending: {'Rahul': 88500, 'Sneha': 76000, 'Arjun': 1500, 'Priya': 12000, 'Karan': 3000}
Top Customer: Rahul
```

```python
# ------------------------------
# PART 5 - CUSTOMER ANALYSIS
# ------------------------------

print("\n--- Customer Analysis ---")

customer_spending = {}

for order in orders:
    name = order["customer"]
    pid = order["product_id"]
    qty = order["quantity"]

    amount = products[pid]["price"] * qty
    customer_spending[name] = customer_spending.get(name, default: 0) + amount

print("Customer Spending:", customer_spending)
```

```
--- Customer Analysis ---
Customer Spending: {'Rahul': 88500, 'Sneha': 76000, 'Arjun': 1500, 'Priya': 12000, 'Karan': 3000}
Top Customer: Rahul
Customers spending > 50000: ['Rahul', 'Sneha']
Product Revenue Tuples: [('Laptop', 150000), ('Mouse', 2500), ('Keyboard', 4500), ('Monitor', 24000)]

--- Final Challenge ---
Visited but never ordered: set()
Ordered but visited once: ['Priya', 'Karan']
```

```python
# ------------------------------
# PART 7 - TUPLE USAGE
# ------------------------------

product_tuples = [(p, product_revenue[p]) for p in product_revenue]
print("Product Revenue Tuples:", product_tuples)


# ------------------------------
# PART 8 - REPORT GENERATION
# ------------------------------

with open("sales_report.txt", "w", encoding="utf-8") as f:
    f.write("E-Commerce Sales Report\n")
    f.write(f"Total Website Visits: {len(visits)}\n")
    f.write(f"Unique Visitors: {len(unique_visitors)}\n")
    f.write(f"Total Revenue: {total_revenue}\n\n")
```


```
--- Customer Analysis ---
Customer Spending: {'Rahul': 88500, 'Sneha': 76000, 'Arjun': 1500, 'Priya': 12000, 'Karan': 3000}
Top Customer: Rahul
Customers spending > 50000: ['Rahul', 'Sneha']
Product Revenue Tuples: [('Laptop', 150000), ('Mouse', 2500), ('Keyboard', 4500), ('Monitor', 24000)]

--- Final Challenge ---
Visited but never ordered: set()
Ordered but visited once: ['Priya', 'Karan']
```

```python
# FINAL CHALLENGE
# ------------------------------

print("\n--- Final Challenge ---")

order_customers = set([o["customer"] for o in orders])

# Visitors who never ordered
never_ordered = unique_visitors - order_customers
print("Visited but never ordered:", never_ordered)

# Customers who visited only once but ordered
one_time_visitors = [v for v in visit_count if visit_count[v] == 1]
special_customers = [c for c in customer_orders if c in one_time_visitors]

print("Ordered but visited once:", special_customers)
```

```
Top Customer: Rahul
Customers spending > 50000: ['Rahul', 'Sneha']
Product Revenue Tuples: [('Laptop', 150000), ('Mouse', 2500), ('Keyboard', 4500), ('Monitor', 24000)]

--- Final Challenge ---
Visited but never ordered: set()
Ordered but visited once: ['Priya', 'Karan']

Process finished with exit code 0
```